Chapter 10 — Electromagnetism

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Force on conductor: (a) BIL cos θ (b) BIL sin θ (c) BI/L (d) BIL tan θ
2. SI unit of flux: (a) farad (b) weber (c) tesla (d) henry
3. Magnetic flux: (a) B/A (b) A/B (c) BA sin θ (d) BA cos θ
4. Radius of charge path: (a) mv/qB (b) q/mvB (c) qB/mv (d) mvB/q
5. Motional emf: (a) v/B L (b) vB/L (c) vBL (d) BL/v
6. Faraday's law: (a) ε = Φ/Nt (b) ε = −NΔΦ/Δt (c) ε = NΦt (d) ε = NΔt/ΔΦ
7. Ferrofluids contain: (a) only water (b) magnetic nanoparticles (c) iron filings (d) only oil
8. Seismometer emf ∝: (a) velocity (b) mass (c) displacement (d) temperature

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is the force on a conductor maximum when perpendicular to the field?
2. Why does a magnetic field change the direction of a charge but not its speed?
3. Why is magnetic flux zero when the field is parallel to the plane of a coil?
4. How does faster motion increase the induced emf?
5. Why does a ferrofluid form spikes in a magnetic field?
6. How does inertia help a seismometer detect ground motion?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Derive the force on a current-carrying conductor in a magnetic field. **(4)**
(b) Define magnetic flux and flux density and explain their relation. **(4)**
Q.4 (a) Describe the motion of a charged particle in a uniform magnetic field and derive r = mv/qB. **(4)**
(b) State Faraday's law and explain the factors affecting induced emf. **(4)**

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